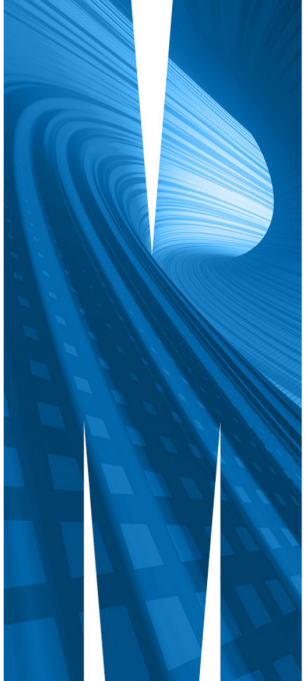
MONASH University

ETC3550/ETC5550
Applied forecasting

Week 11: Regression models



Multiple regression and forecasting

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \cdots + \beta_k x_{k,t} + \varepsilon_t.$$

- y_t is the variable we want to predict: the "response" variable
- Each $x_{j,t}$ is numerical and is called a "predictor". They are usually assumed to be known for all past and future times.
- The coefficients $\beta_1, \ldots, \beta_k$ measure the effect of each predictor *after taking account of the effect of all other predictors in the model*.
- ε_t is a white noise error term

Uses of dummy variables

Seasonal dummies

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- For monthly data: use 11 dummies
- For daily data: use 6 dummies
- What to do with weekly data?

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Public holidays

- For daily data: if it is a public holiday, dummy=1, otherwise dummy=0.

Holidays

For monthly data

- Christmas: always in December so part of monthly seasonal effect
- Easter: use a dummy variable $v_t = 1$ if any part of Easter is in that month, $v_t = 0$ otherwise.
- Ramadan and Chinese New Year similar.

Distributed lags

Lagged values of a predictor.

Example: x is advertising which has a delayed effect

x_1 = advertising for previous month;
x_2 = advertising for two months previously;
$\vdots$
x_m = advertising for m months previously.

Fourier series

Periodic seasonality can be handled using pairs of Fourier terms:

$$s_\ell(t) = \sin\left(\frac{2\pi\ell t}{m}\right) \qquad c_\ell(t) = \cos\left(\frac{2\pi\ell t}{m}\right)$$

$$y_t = a + bt + \sum_{\ell=1}^{K}\left[\alpha_\ell s_\ell(t) + \beta_\ell c_\ell(t)\right] + \varepsilon_t$$

- Every periodic function can be approximated by sums of sin and cos terms for large enough K.
- Choose K by minimizing AICc or CV.
- Called “harmonic regression”

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Maximizing $\bar{R}^2$ is equivalent to minimizing $\hat{\sigma}^2$.

$$\hat{\sigma}^2 = \frac{1}{T-k-1}\sum_{t=1}^{T}\varepsilon_t^2$$

Akaike's Information Criterion

$$\text{AIC} = -2\log(L) + 2(k+2)$$

- L = likelihood
- k = # predictors in model.
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$$\text{AIC}_C = \text{AIC} + \frac{2(k+2)(k+3)}{T-k-3}$$

- Minimizing the AIC or AICc is asymptotically equivalent to minimizing MSE via **leave-one-out cross-validation** (for any linear regression).

Leave-one-out cross-validation

For regression, leave-one-out cross-validation is faster and more efficient than time-series cross-validation.

- Select one observation for test set, and use *remaining* observations in training set. Compute error on test observation.
- Repeat using each possible observation as the test set.
- Compute accuracy measure over all errors.

Cross-validation

Traditional evaluation

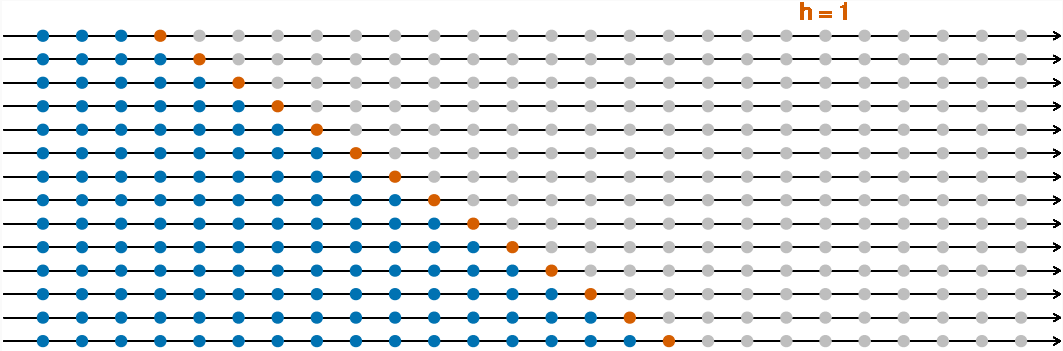


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Time series cross-validation

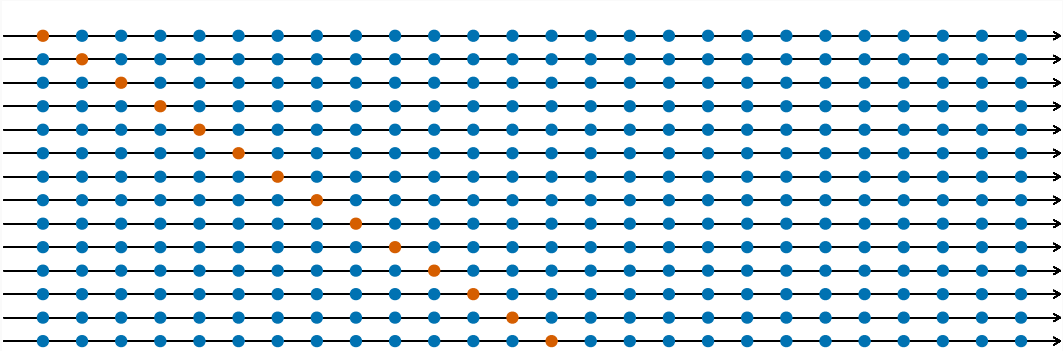


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Leave-one-out cross-validation

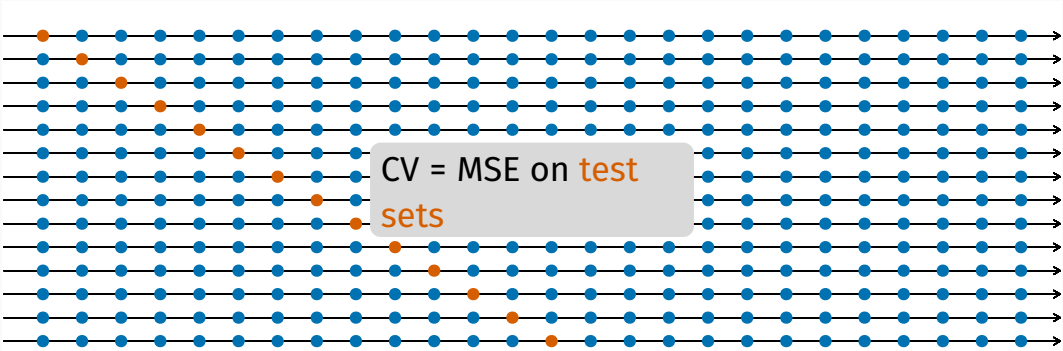


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Choosing regression variables

Best subsets regression

- Fit all possible regression models using one or more of the predictors.
- Choose the best model based on one of the measures of predictive ability (CV, AIC, AICc).

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Backwards stepwise regression

- Start with a model containing all variables.
- Subtract one variable at a time. Keep model if lower CV.
- Iterate until no further improvement.
- Not guaranteed to lead to best model.

Ex-ante versus ex-post forecasts

- *Ex ante forecasts* are made using only information available in advance.
  - require forecasts of predictors
- *Ex post forecasts* are made using later information on the predictors.
  - useful for studying behaviour of forecasting models.
- trend, seasonal and calendar variables are all known in advance, so these don't need to be forecast.